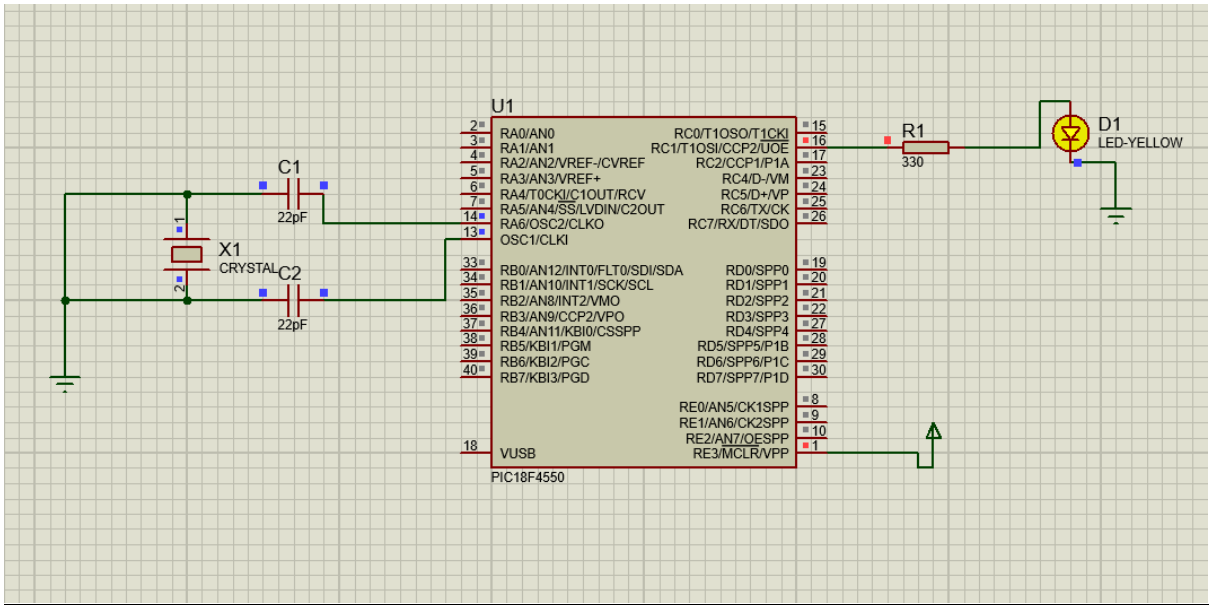
# Practical No:7

## Circuit:




## Code:

```
#include <xc.h>
#define _XTAL_FREQ 8000000
#pragma config FOSC = HS
#pragma config WDT = OFF
#pragma config LVP = OFF
#pragma config PBADEN = OFF
#pragma config MCLRE = ON
#pragma config XINST = OFF

void delay_1ms(void)
{
   T0CON = 0x08;
   TMR0H = 0xF8;
   TMR0L = 0x30;

   INTCONbits.TMR0IF = 0;
   T0CONbits.TMR0ON = 1;
```

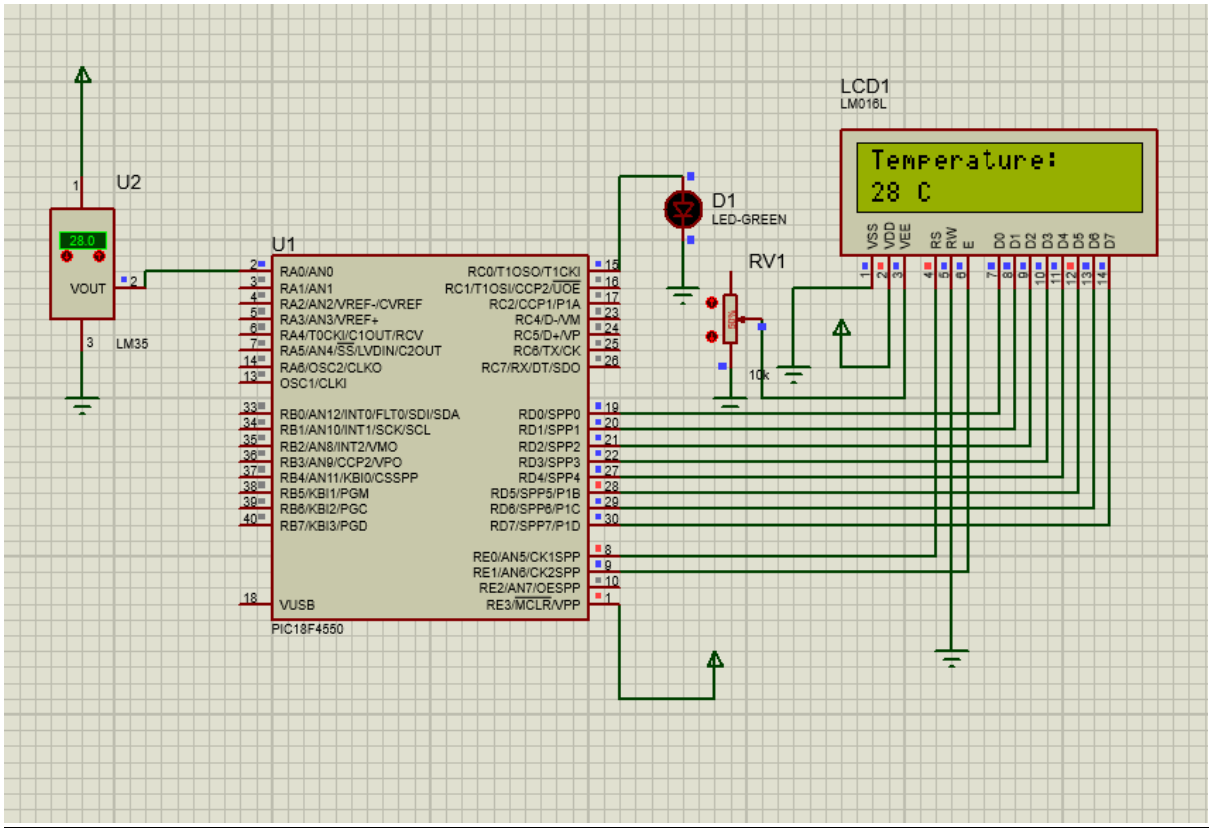
```
   while (INTCONbits.TMR0IF == 0);
   T0CONbits.TMR0ON = 0;
   INTCONbits.TMR0IF = 0;
}
void delay_1s(void)
{
   unsigned int i;
 for (i = 0; i < 1000; i++)
   {
      delay_1ms();
   }
}

void main(void)
{
   ADCON1 = 0x0F;
   TRISCbits.TRISC1 = 0;
   LATCbits.LATC1 = 0;
while (1)
   {
      LATCbits.LATC1 = 1;
      delay_1s();

      LATCbits.LATC1 = 0;
      delay_1s();
   }
}
```

# Practical No:8

## Circuit:




## Code:

```
#include <xc.h>
#include <stdio.h>
#define _XTAL_FREQ 8000000
#pragma config FOSC = HS
#pragma config WDT = OFF
#pragma config LVP = OFF
#pragma config PBADEN = OFF
#pragma config MCLRE = ON
#pragma config XINST = OFF
#define RS LATEbits.LATE0
#define EN LATEbits.LATE1

char text[17];
```

```
void LCD_Command(unsigned char cmd)
{
   LATD = cmd;
   RS = 0;
   EN = 1;
   __delay_ms(2);
   EN = 0;
   __delay_ms(2);
}
void LCD_Char(unsigned char data)
{
   LATD = data;
   RS = 1;
   EN = 1;
   __delay_ms(2);
   EN = 0;
   __delay_ms(2);
}
void LCD_String(char *s)
{
   while (*s)
      LCD_Char(*s++);
}
void LCD_Init(void)
{
   TRISD = 0x00;
   TRISEbits.TRISE0 = 0;
   TRISEbits.TRISE1 = 0;
   LATD = 0;
   LATE = 0;
   __delay_ms(20);
   LCD_Command(0x38);
   LCD_Command(0x0C);
   LCD_Command(0x06);
```

```
  LCD_Command(0x01);
   __delay_ms(5);
}
void main(void)
{
   unsigned int adc;
   unsigned int temperature;
   ADCON1 = 0x0E;
   ADCON2 = 0xAE;
   ADCON0 = 0x01;
   TRISAbits.TRISA0 = 1;
   LCD_Init();
   while (1)
   {
      ADCON0 = 0x01;
      __delay_us(25);

      ADCON0bits.GO = 1;
      while (ADCON0bits.GO);

      adc = ((unsigned int)ADRESH << 8) | ADRESL;

      temperature = (adc * 500UL) / 1023;

      LCD_Command(0x80);
      LCD_String("Temperature:");

      LCD_Command(0xC0);
      sprintf(text, "%u C        ", temperature);
      LCD_String(text);

      __delay_ms(500);
   }
}
```